

DML Task

-- INSERT INTO Department

```
INSERT INTO Department VALUES (10, 'IT', NULL, NULL);
```

```
INSERT INTO Department VALUES (20, 'HR', NULL, NULL);
```

```
INSERT INTO Department VALUES (30, 'Finance', NULL, NULL);
```

-- INSERT INTO Employee

```
INSERT INTO Employee VALUES (1001, 'Ali', 'Ahmed', DATE '1998-05-10', 'Male',  
NULL, 10);
```

```
INSERT INTO Employee VALUES (1002, 'Sara', 'Khalid', DATE '1999-03-15',  
'Female', 1001, 10);
```

```
INSERT INTO Employee VALUES (1003, 'Omar', 'Said', DATE '1997-07-20', 'Male',  
1001, 20);
```

```
INSERT INTO Employee VALUES (1004, 'Mona', 'Salim', DATE '2000-01-25',  
'Female', 1003, 30);
```

```
INSERT INTO Employee VALUES (1005, 'Nasser', 'Hamad', DATE '1996-11-12',  
'Male', 1001, 10);
```

-- UPDATE Department managers

```
UPDATE Department
```

```
SET MGR_SSN = 1001, HIRING_DATE = DATE '2024-01-01' WHERE DNUM = 10;
```

```
UPDATE Department
```

```
SET MGR_SSN = 1003, HIRING_DATE = DATE '2024-02-01'
```

```
WHERE DNUM = 20;
```

```
UPDATE Department
```

```
SET MGR_SSN = 1004, HIRING_DATE = DATE '2024-03-01'
```

```
WHERE DNUM = 30;
```

-- INSERT INTO Dept_Loc

INSERT INTO Dept_Loc VALUES (10, 'Muscat');

INSERT INTO Dept_Loc VALUES (10, 'Seeb');

INSERT INTO Dept_Loc VALUES (20, 'Sohar');

INSERT INTO Dept_Loc VALUES (30, 'Nizwa');

-- INSERT INTO Project

INSERT INTO Project VALUES (501, 'Website System', 'Muscat', 'Muscat', 10);

INSERT INTO Project VALUES (502, 'Payroll System', 'Sohar', 'Sohar', 20);

INSERT INTO Project VALUES (503, 'Budget System', 'Nizwa', 'Nizwa', 30);

-- INSERT INTO Dependent

INSERT INTO Dependent VALUES ('Hana', 1001, DATE '2015-04-10', 'Female');

INSERT INTO Dependent VALUES ('Yousef', 1002, DATE '2018-09-12', 'Male');

INSERT INTO Dependent VALUES ('Laila', 1003, DATE '2016-06-05', 'Female');

-- INSERT INTO Working_Hours

INSERT INTO Working_Hours VALUES (501, 1001, 20);

INSERT INTO Working_Hours VALUES (501, 1002, 15);

INSERT INTO Working_Hours VALUES (502, 1003, 25);

INSERT INTO Working_Hours VALUES (503, 1004, 30);

INSERT INTO Working_Hours VALUES (501, 1005, 10);

COMMIT;

-- UPDATE examples

```
UPDATE Employee  
SET LNAME = 'Al-Habsi'  
WHERE SSN = 1002;
```

```
UPDATE Project  
SET CITY = 'Muscat'  
WHERE PNUM = 502;
```

```
UPDATE Working_Hours  
SET HOURS = 18  
WHERE SSN = 1002 AND PNUM = 501;
```

-- DELETE examples

```
DELETE FROM Working_Hours  
WHERE SSN = 1005 AND PNUM = 501;
```

```
DELETE FROM Dependent  
WHERE SSN = 1003 AND DEPNAME = 'Laila';  
COMMIT;
```

--TO MAKE SURE

```
SELECT * FROM Department;  
SELECT * FROM Employee;  
SELECT * FROM Project;  
SELECT * FROM Dependent;  
SELECT * FROM Dept_Loc;  
SELECT * FROM Working_Hours;
```

Query Results Screenshots:

Script Output x Query Result x Query Result 1 x Query Result 2 x Query Result 3 x Query Result 4 x Query Result 5 x

SQL | All Rows Fetched: 3 in 0.03 seconds

	DNUM	DNAME	MGR_SSN	HIRING_DATE
1	10	IT	1001	01-JAN-24
2	20	HR	1003	01-FEB-24
3	30	Finance	1004	01-MAR-24

| Line 140 Column 29 | Insert | Modified | Windows: C

Script Output x Query Result x Query Result 1 x Query Result 2 x Query Result 3 x Query Result 4 x Query Result 5 x

SQL | All Rows Fetched: 5 in 0.007 seconds

	SSN	FNAME	LNAME	DOB	GENDER	SUPER_SSN	DNUM
1	1001	Ali	Ahmed	10-MAY-98	Male	(null)	10
2	1002	Sara	Al-Habsi	15-MAR-99	Female	1001	10
3	1003	Omar	Said	20-JUL-97	Male	1001	20
4	1004	Mona	Salim	25-JAN-00	Female	1003	30
5	1005	Nasser	Hamad	12-NOV-96	Male	1001	10

| Line 140

Script Output x Query Result x Query Result 1 x Query Result 2 x Query Result 3 x Query Result 4 x Query Result 5 x

SQL | All Rows Fetched: 3 in 0.005 seconds

	PNUM	PNAME	LOCATION	CITY	DNUM
1	501	Website System	Muscat	Muscat	10
2	502	Payroll System	Sohar	Muscat	20
3	503	Budget System	Nizwa	Nizwa	30

Script Output x Query Result x Query Result 1 x Query Result 2 x Query Result 3 x Query Result 4 x Query Result 5 x

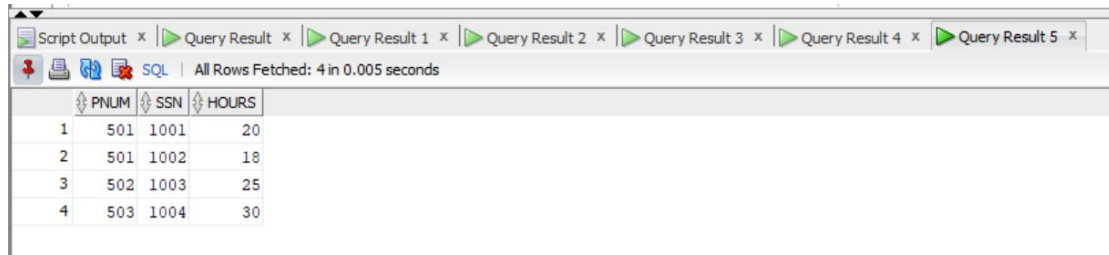
SQL | All Rows Fetched: 2 in 0.007 seconds

	DEPNAME	SSN	DOB	GENDER
1	Hana	1001	10-APR-15	Female
2	Yousef	1002	12-SEP-18	Male

Script Output x Query Result x Query Result 1 x Query Result 2 x Query Result 3 x Query Result 4 x Query Result 5 x

SQL | All Rows Fetched: 4 in 0.007 seconds

	DNUM	LOCATION
1	10	Muscat
2	10	Seeb
3	20	Sohar
4	30	Nizwa



The screenshot shows a SQL query result window with a table containing 4 rows and 3 columns: PNUM, SSN, and HOURS. The table data is as follows:

	PNUM	SSN	HOURS
1	501	1001	20
2	501	1002	18
3	502	1003	25
4	503	1004	30

Note about Errors Log:

No execution errors were encountered during DML implementation.

All INSERT, UPDATE, and DELETE statements executed successfully.